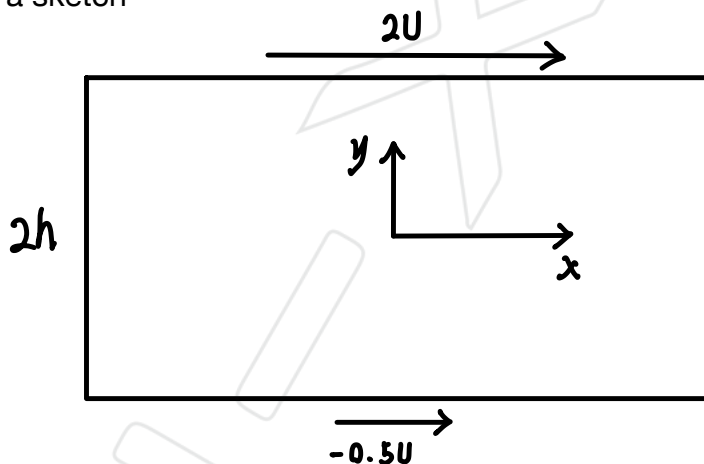


2018 Question 3:

3. A two-dimensional, parallel-sided channel of height $2h$ has its upper wall moving with velocity $2U$ and its lower wall moving with velocity $-0.5U$. A flow in the channel is caused solely by the movement of the walls and no pressure gradient is applied.

- What assumptions are made in a fully-developed, incompressible, steady laminar channel flow about the behaviour of the velocity?
- Starting from the equations of motion for steady viscous flow, derive an expression for the velocity profile across the duct.
- Calculate the vorticity profile.
- The lower wall is now brought to rest. Determine the change in shear stress at the wall after steady flow conditions have been re-established.

Start with a sketch



(a) Fully developed flow: $\frac{\partial u}{\partial x} = 0$

$$\text{In 2D, } \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad \therefore \frac{\partial v}{\partial y} = 0 \quad \therefore \text{velocity } v \text{ constant}$$

$$\text{But } v(y=0) = 0 \rightarrow v = 0 \text{ everywhere}$$

$$\rightarrow \frac{\partial u}{\partial x} = 0 \text{ everywhere}$$

(b) Equations of motion for steady flow (u-component):

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{dp}{dx} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

$$\therefore 0 = -\frac{1}{\rho} \frac{dp}{dx} + \nu \frac{\partial^2 u}{\partial y^2}$$

$$\therefore \frac{dp}{dx} = 0 \quad \therefore \frac{\partial^2 u}{\partial y^2} = 0$$

$$\text{Integrate twice} \quad \therefore \frac{\partial u}{\partial y} = a \quad \therefore \partial u = a \partial y \quad \therefore u = ay + b$$

Apply no-slip boundary condition

$$u(h) = 2U \longrightarrow 2U = ah + b$$

$$u(-h) = -0.5U \longrightarrow -0.5U = a(-h) + b$$

$$\therefore a = \frac{5U}{4h}, \quad b = \frac{3U}{4}$$

$$\therefore u(y) = \frac{U}{4} \left(5 \frac{y}{h} + 3 \right)$$

(c) Recalling the definition of Vorticity:

$$\omega_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$$

$$\therefore \gamma = 0$$

$$\therefore \frac{\partial v}{\partial x} = 0$$

$$\therefore \omega_z = -\frac{\partial u}{\partial y} = -\frac{5U}{4h}$$

(d) Shear stress at lower wall when $u = -0.5U$

$$\tau_w = \mu \left. \frac{du}{dy} \right|_{y=-h} = \mu \frac{5U}{4h}$$

Lower wall is now $u = 0$.

$$u = ay + b$$

$$\therefore \begin{cases} 0 = -ah + b \\ 2U = ah + b \end{cases} \quad \therefore \begin{cases} a = U \\ b = \frac{U}{h} \end{cases}$$

$$\therefore u(y) = \frac{U}{h}y + U$$

$$\therefore \tau_{w(\text{new})} = \mu \frac{\partial u}{\partial y} = \mu \frac{U}{h}$$

$$\therefore \Delta \tau_w = \tau_w - \tau_{w(\text{new})}$$

$$= \frac{1}{4} \mu \frac{U}{h}$$

2019 Question 3:

- (a) Sketch the velocity profile, in the form y/δ versus u/U_e , for a boundary layer subject to a zero pressure gradient. Also sketch the corresponding profile for a boundary layer in adverse pressure gradient.
- (b) Explain why the shape of the velocity profile close to the wall depends on the pressure gradient.
- (c) A boundary layer profile is given by:

$$\frac{u}{U_e} = 3\left(\frac{y}{\delta}\right)^2 - 2\left(\frac{y}{\delta}\right)^3.$$

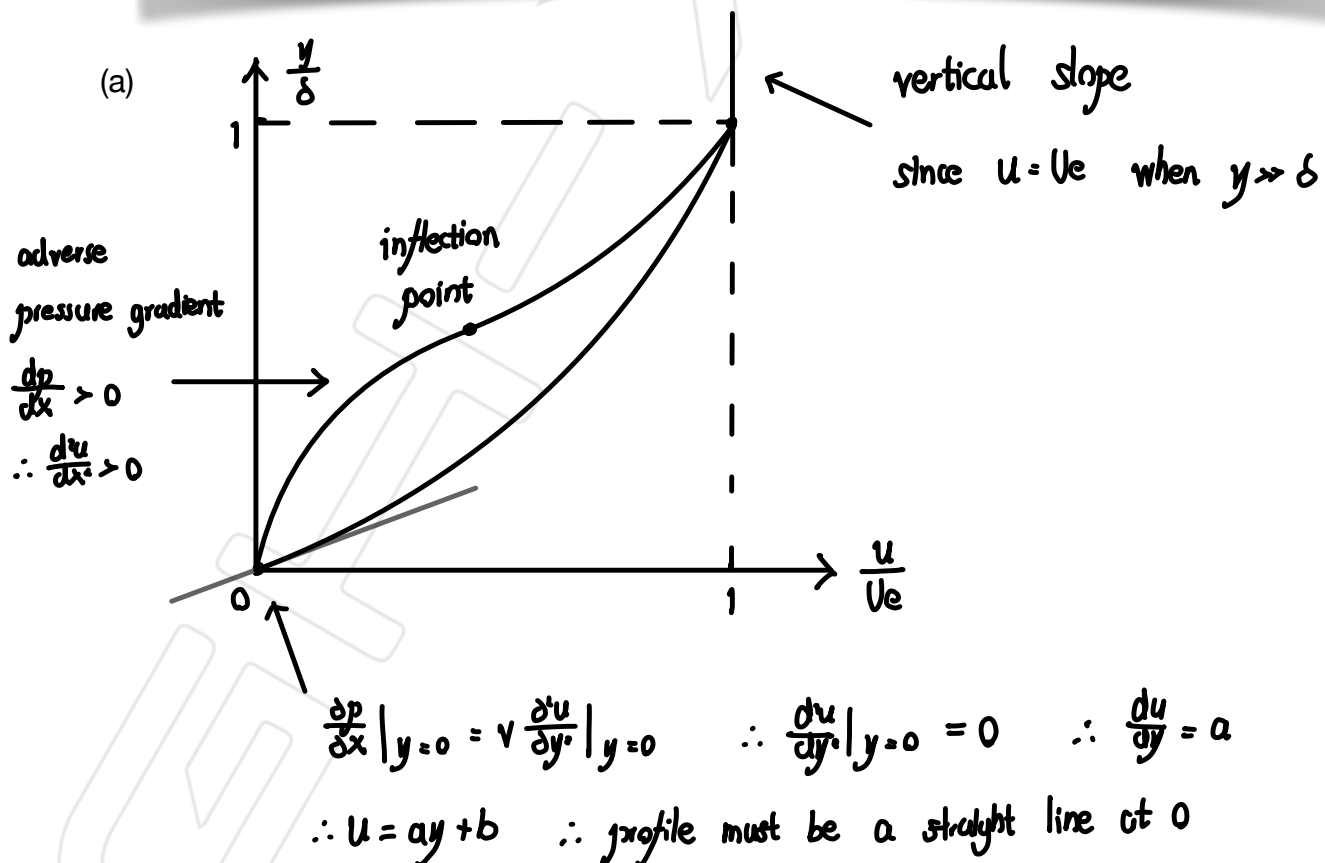
Show that the boundary layer is on the verge of separation.

- (d) The mass flow deficit of a boundary layer (the "displacement thickness", δ^*) is defined as:

$$\frac{\delta^*}{\delta} = \int_0^1 \left(1 - \frac{u}{U_e}\right) d(y/\delta).$$

Determine its value for the given profile.

- (e) On your sketch from part (a), shade the area corresponding to the displacement thickness. Explain what it means and why your estimate of δ^* is large in this case.



(b) At wall:

~~$$u \frac{\partial u}{\partial x} \Big|_w + v \frac{\partial u}{\partial y} \Big|_w = -\frac{1}{\rho} \frac{\partial p}{\partial x} \Big|_w + \nu \frac{\partial^2 u}{\partial x^2} \Big|_w + \nu \frac{\partial^2 u}{\partial y^2} \Big|_w$$~~

$$\therefore \frac{\partial^2 u}{\partial y^2} \Big|_{\text{wall}} \propto \frac{\partial p}{\partial x} \Big|_{\text{wall}}$$

$$(c) \frac{u}{U_e} = 3\left(\frac{y}{\delta}\right)^2 - 2\left(\frac{y}{\delta}\right)^3$$

At the point of separation, $\tau_w = 0$

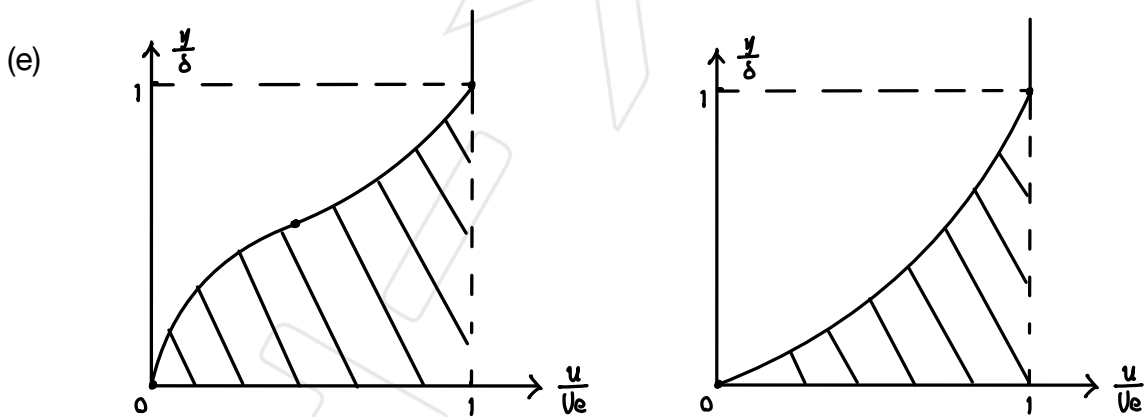
$$\therefore \frac{\partial \left(\frac{u}{U_e}\right)}{\partial y} = \frac{6}{\delta} \left(\frac{y}{\delta}\right) - \frac{6}{\delta} \left(\frac{y}{\delta}\right)^2$$

$$\text{So at } y=0, \frac{\partial \left(\frac{u}{U_e}\right)}{\partial y} = 0$$

$$\therefore \frac{\partial u}{\partial y} = 0$$

$$\therefore \tau_w = 0$$

$$\begin{aligned}
 \text{(d)} \quad \frac{\delta^*}{\delta} &= \int_0^1 \left(1 - \frac{u}{u_e}\right) d\frac{y}{\delta} \\
 &= \int_0^1 \left(1 - 3\left(\frac{y}{\delta}\right)^2 + 2\left(\frac{y}{\delta}\right)^3\right) d\frac{y}{\delta} \\
 &= \left[\frac{y}{\delta} - \left(\frac{y}{\delta}\right)^3 + \frac{1}{2}\left(\frac{y}{\delta}\right)^4\right]_0^1 \\
 &= 1 - 1 + \frac{1}{2} \\
 &= \frac{1}{2}
 \end{aligned}$$



The thickness that we must displace the wall vertically to generate an inviscid flow (slip condition) without boundary layer